King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 201 Final Exam Term 233

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

---

1. The area of the parallelogram that has the vectors

$$\vec{u} = \langle 1, -2, 3 \rangle, \quad \vec{v} = \langle 2, -1, 0 \rangle$$

as adjacent sides is equal to

(a) $5\sqrt{6}$

(b) $2\sqrt{6}$

(c) $6\sqrt{6}$

(d) $3\sqrt{6}$

(e) $4\sqrt{6}$

2. The volume of the solid bounded above by the sphere $x^2 + y^2 + z^2 = 4$ and below by the cone $z^2 = 3x^2 + 3y^2$ is equal to

(a) $\dfrac{8\pi}{3}$

(b) $\dfrac{16\pi}{3}(\sqrt{3} - 1)$

(c) $\dfrac{8\pi}{3}\left(1 - \dfrac{\sqrt{3}}{2}\right)$

(d) $\dfrac{4\pi}{3}(3 - \sqrt{3})$

(e) $\dfrac{8\pi}{3}(2 - \sqrt{3})$

3. The volume of the solid bounded by the paraboloids $z = x^2 + y^2 + 2$ and $z = 8 - x^2 - y^2$ is equal to

(a) 9π

(b) $2\pi\sqrt{3}$

(c) 6π

(d) $4\pi\sqrt{3}$

(e) $\dfrac{5\pi}{2}$

4. Let $d > 0$. If the volume of the solid bounded by the planes

$$x + 2y + z = d, \quad x = 0, \quad y = 0, \quad z = 0$$

is equal to $\dfrac{16}{3}$, then $d =$

(a) 4

(b) 2

(c) 3

(d) 5

(e) 6

5. $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{3/2}\, dy\, dx =$

(a) $\dfrac{3\pi}{10}$

(b) π

(c) $\dfrac{\pi}{5}$

(d) $\dfrac{2\pi}{5}$

(e) $\dfrac{\pi}{10}$

6. If (a, b) is the only critical point of $f(x, y) = x^2 - xy - y^2 - 3x - y$, then

(a) f has a saddle point at (a, b)

(b) $a + b = 3$

(c) $ab = -2$

(d) f has a relative minimum at (a, b)

(e) f has a relative maximum at (a, b)

7. $\lim_{(x,y)\to(0,0)} \dfrac{1 - \sqrt{1 - x^2 - y^2}}{x^2 + y^2} =$

(a) does not exist

(b) $\dfrac{1}{2}$

(c) $-\dfrac{1}{4}$

(d) 0

(e) -2

8. If $f(x,y) = \dfrac{y+x}{y+1}$, then $\nabla f(0,1) =$

(a) $\dfrac{1}{2}\vec{i}$

(b) $-\dfrac{1}{2}\vec{j}$

(c) $\vec{i} - \dfrac{1}{4}\vec{j}$

(d) $\dfrac{1}{2}\vec{i} + \dfrac{1}{4}\vec{j}$

(e) $2\vec{i} + 4\vec{j}$

9. The function $f(x,y,z) = \dfrac{\ln(1-z)}{e^x - e^y}$ is continuous on

(a) $\{(x,y,z) : x \neq y,\ z < 1\}$

(b) $\{(x,y,z) : x \neq y,\ z > 1\}$

(c) $\{(x,y,z) : x > y,\ z > 1\}$

(d) $\{(x,y,z) : x < y,\ z < 1\}$

(e) $\{(x,y,z) : x \neq y,\ z \leq 1\}$

10. If $ax + by - z = d$ is an equation of the tangent plane to the paraboloid $z = x^2 + 2y^2 + 1$ at the point $(2,1,7)$, then $a + b + d =$

(a) 10

(b) 13

(c) 8

(d) -6

(e) -7

11. An equation for the tangent line to the parametric curve

$$x = 1 + 2\sin t, \quad y = t - \cos^2 t$$

at the point corresponding to $t = 0$ is

(a) $y = \dfrac{1}{2}x - \dfrac{3}{2}$

(b) $y = \dfrac{1}{2}x + \dfrac{3}{2}$

(c) $y = -2x + 1$

(d) $y = 2x - 3$

(e) $y = -\dfrac{1}{2}x - \dfrac{1}{2}$

12. The average value of $f(x, y) = x^2 + y^2$ over the square region with vertices $(0, 0)$, $(0, 2)$, $(2, 0)$, $(2, 2)$ is equal to

(a) $\dfrac{16}{3}$

(b) $\dfrac{8}{3}$

(c) $\dfrac{20}{3}$

(d) $\dfrac{10}{3}$

(e) 5

13. The directional derivative of $f(x, y) = xy$ at $P(2, -1)$ in the direction of the vector $\vec{v} = \langle 1, 1 \rangle$ is equal to

(a) $\dfrac{\sqrt{2}}{3}$

(b) 2

(c) $\dfrac{\sqrt{2}}{2}$

(d) $2\sqrt{2}$

(e) $\sqrt{2}$

14. $\int_0^{\pi/2} \int_0^{\cos x} (1 + \sin x)\, dy\, dx =$

(a) $\dfrac{1}{2}$

(b) 1

(c) $\dfrac{5}{2}$

(d) 2

(e) $\dfrac{3}{2}$

15. The maximum value of $f(x, y) = 2x + 2xy + y$ subject to the constraint $2x + y = 20$ is

(a) 60

(b) 80

(c) 140

(d) 120

(e) 100

16. The point with spherical coordinates $(\rho, \theta, \phi) = \left(4, \frac{\pi}{6}, \frac{\pi}{4}\right)$ is represented in rectangular coordinates by

(a) $(x, y, z) = (\sqrt{6}, 2\sqrt{2}, 1)$

(b) $(x, y, z) = (2\sqrt{6}, \sqrt{2}, \sqrt{2})$

(c) $(x, y, z) = (\sqrt{6}, \sqrt{2}, 2\sqrt{2})$

(d) $(x, y, z) = (2\sqrt{6}, 2\sqrt{2}, 2\sqrt{2})$

(e) $(x, y, z) = (\sqrt{6}, \sqrt{2}, \sqrt{2})$

17. The minimum distance from the point $(-1, -2, 0)$ to the surface $z = \sqrt{1 - 2x - 2y}$ is equal to (Hint: to simplify the computations, minimize the square of the distance)

(a) $2\sqrt{7}$

(b) $\sqrt{5}$

(c) $2\sqrt{5}$

(d) $\frac{\sqrt{5}}{2}$

(e) $\sqrt{7}$

18. The area of the region inside the circle $r = 3\sin\theta$ for $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{3}$ is equal to

(a) $\frac{5\pi}{4}$

(b) $\frac{3\pi}{8}$

(c) $\frac{\pi}{6}$

(d) $\frac{5\pi}{8}$

(e) $\frac{2\pi}{5}$

19. $\int_0^4 \int_{\sqrt{y}}^2 f(x, y)\, dx\, dy =$

(a) $\int_0^2 \int_0^{\sqrt{x}} f(x, y)\, dy\, dx$

(b) $\int_0^4 \int_{\sqrt{x}}^2 f(x, y)\, dy\, dx$

(c) $\int_0^2 \int_{x^2}^0 f(x, y)\, dy\, dx$

(d) $\int_{\sqrt{y}}^2 \int_0^4 f(x, y)\, dy\, dx$

(e) $\int_0^2 \int_0^{x^2} f(x, y)\, dy\, dx$

20. An equation for the plane that passes through the point $(1, 2, 3)$ and is parallel to the yz-plane is

(a) $z = 3$

(b) $y = 2$

(c) $x + y = 3$

(d) $x = 1$

(e) $y + z = 5$

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(d)**	$\vec{u} \times \vec{v} = \langle 3, 6, 3 \rangle$, length $3\sqrt{6}$.
2	**(e)**	$\phi \le \frac{\pi}{6}$: $V = 2\pi \cdot \frac{8}{3}(1 - \frac{\sqrt{3}}{2}) = \frac{8\pi}{3}(2 - \sqrt{3})$.
3	**(a)**	$r = \sqrt{3}$: $V = \int_0^{2\pi} \int_0^{\sqrt{3}} (6 - 2r^2) r \, dr \, d\theta = 9\pi$.
4	**(a)**	$V = \frac{d^3}{12} = \frac{16}{3} \Rightarrow d = 4$.
5	**(e)**	Polar: $\int_0^{\pi/2} \int_0^1 r^4 \, dr \, d\theta = \frac{\pi}{10}$.
6	**(a)**	Critical point $(1, -1)$, $D = -5 < 0$: saddle point.
7	**(b)**	Conjugate gives $\frac{1}{1+\sqrt{1-x^2-y^2}} \to \frac{1}{2}$.
8	**(d)**	$\nabla f = \langle \frac{1}{y+1}, \frac{1-x}{(y+1)^2} \rangle$, at $(0, 1)$: $\langle \frac{1}{2}, \frac{1}{4} \rangle$.
9	**(a)**	Need $1 - z > 0$ and $e^x \neq e^y$: $x \neq y$, $z < 1$.
10	**(b)**	Plane $4x + 4y - z = 5$, so $a + b + d = 13$.
11	**(a)**	$(1, -1)$, $m = \frac{1}{2}$: $y = \frac{1}{2}x - \frac{3}{2}$.
12	**(b)**	Integral $= \frac{32}{3}$, area 4: average $\frac{8}{3}$.
13	**(c)**	$\nabla f = \langle -1, 2 \rangle$, $\vec{u} = \frac{1}{\sqrt{2}}\langle 1, 1 \rangle$: $\frac{\sqrt{2}}{2}$.
14	**(e)**	Inner gives $(1 + \sin x)\cos x$; $u = 1 + \sin x$: $\int_1^2 u \, du = \frac{3}{2}$.
15	**(d)**	$y = 2x$, $x = 5, y = 10$: $f = 120$ (bigger than $f(0, 20) = 20$).
16	**(c)**	$(\sqrt{6}, \sqrt{2}, 2\sqrt{2})$.
17	**(b)**	Minimize $(x+1)^2 + (y+2)^2 + 1 - 2x - 2y$: at $(0, -1)$, $d^2 = 5$.
18	**(b)**	$\frac{1}{2} \int 9 \sin^2 \theta \, d\theta = \frac{9}{4} \cdot \frac{\pi}{6} = \frac{3\pi}{8}$.
19	**(e)**	Region $0 \le x \le 2$, $0 \le y \le x^2$.
20	**(d)**	Normal $\langle 1, 0, 0 \rangle$ through $(1, 2, 3)$: $x = 1$.